

Sebastian Ashcallay Silva

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EDUCATION

University of Maryland

Bachelor of Science in Computer Science (ML Concentration), Minor in Mathematics

College Park, MD

Jan. 2024 – Dec 2025

- **GPA:** 3.5 (Dean's List)
- **Relevant Courses:** Machine Learning, Artificial Intelligence, Computer Vision, Data Science, Algorithms, Linear Algebra

EXPERIENCE

Software Developer Intern

Jan. 2025 – Jul. 2025

Apollonian AI

Remote

- Designed and implemented an end-to-end Python pipeline for avatar-defined informational video generation, from prototype to production deployment in a Firebase application used by development teams and selected pilot users
- Built a multimodal AI orchestration layer using the OpenAI Python SDK to coordinate script and image generation, text-to-speech audio and captioning into a single automated pipeline
- Engineered a concurrent, serverless execution model using Firebase Functions on Google Cloud, reducing end-to-end runtime from 5-6 minutes to 2-3 minutes
- Automated video assembly with FFMPEG for caption overlay, audio mixing, scene stitching, and MP4 export, enabling fully unattended batch generation
- Designed and integrated backend data schemas with Firebase Cloud Storage and Firestore Database to store generated videos, assets, metadata, and user interaction signals (likes, comments, tags)
- Modeled the full production workflow in Figma to document pipeline stages, data flow, and failure points, improving cross-team communication between backend and frontend developers
- Built an internal documentation and testing portal using HTML/CSS/JS and deployed on Netlify to validate video generation, retrieval features, and randomized input workflows
- Coordinated early-stage team operations, led daily 30-minute meetings, and planned bi-weekly sprints in Jira

Tutor

Sep. 2024 – Jul. 2025

C2 Education

Bethesda, MD

- Conducted 2-hour weekly tutoring sessions for 1-3 students, covering SAT/ACT/IB/AP exams and STEM courses
- Boosted SAT Math scores from 580-620s into the 700s; reinforced high school Math and CS grades to straight A's
- Expanded responsibilities to supervise middle and elementary school instruction, delivering personalized lessons to younger students with varying attendance in English, Math and Social Studies

PROJECTS

Stereographic Projections | *Python, PyTorch, ResNet18, NumPy, Matplotlib*

Mar. 2025 – May 2025

- Built and fine-tuned a ResNet18 CNN to classify 20 U.S. cities from stellar sky projections, achieving 95.25% precision and 100% Top-3 accuracy
- Created 2 synthetic datasets (2700+ images) with custom augmentation (clouds, blur, light pollution), significantly improving model generalization
- Implemented full training and evaluation pipeline including preprocessing, normalization, transfer learning, and performance analysis with confusion matrices, F1-scores, and precision-recall curves
- Co-authored a detailed 5-page report demonstrating CNN's potential for celestial navigation and spatial inference

Professor Ratings Predictor | *Python, Pandas, Scikit-Learn, NumPy, Matplotlib*

Feb. 2025 – Apr. 2025

- Developed predictive regression models on 37,000+ PlanetTerp university reviews covering 4,100+ professors
- Engineered advanced features with VADER sentiment analysis, one-hot keyword encoding, and Principal Component Analysis (PCA) for dimensionality reduction
- Trained and validated models with 10-fold cross-validation, achieving a Mean R^2 of 0.785 and Mean RMSE of 0.452 using Ridge Regression; compared against SVR and Gradient Boosting.

TECHNICAL SKILLS

Languages: Python, Java, C/C++, SQL, JavaScript, HTML/CSS, Dart, OCaml, Bash

ML/Data: PyTorch, Scikit-Learn, Pandas, NumPy, Matplotlib

Backend/Systems: FastAPI, Node.js, Express.js, Postgres, MongoDB, Redis, Firebase (Google Cloud), FFMPEG

Frontend/Mobile/Developer Tools: React, Flutter, Android Studio, Git, VS Code, PyCharm, Eclipse, Jira, Figma